

## Project Initialization and Planning Phase

|               |                                 |
|---------------|---------------------------------|
| Date          | 15 July 2026                    |
| Team ID       | SWTID-2026-8239                 |
| Project Title | Credit Card Approval Prediction |
| Maximum Marks | 5 Marks                         |

### Project Proposal (Proposed Solution) report

The proposal report aims to develop an intelligent **Credit Card Approval Prediction System** using Machine Learning that predicts whether an applicant is eligible for a credit card based on personal, financial, and employment details, enabling faster and more consistent decision-making. Key features include a machine learning-based credit model and real-time decision-making.

| Project Overview  |                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | The primary objective is to develop an intelligent <b>Credit Card Approval Prediction System</b> using Machine Learning that predicts whether an applicant is eligible for a credit card based on personal, financial, and employment details, enabling faster and more consistent decision-making.                                                                                                              |
| Scope             | The project focuses on automating the credit card approval process by analyzing applicant information, training a Random Forest classification model, and deploying it as a user-friendly Flask web application for real-time prediction.                                                                                                                                                                        |
| Problem Statement |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Description       | Traditional credit card approval processes are often time-consuming, require extensive manual verification, and may lead to inconsistent decisions. An automated prediction system is needed to improve efficiency and support data-driven approval decisions.                                                                                                                                                   |
| Impact            | Implementing the proposed system will reduce processing time, improve consistency in approval decisions, assist financial institutions in applicant screening, and enhance the overall customer experience through faster responses.                                                                                                                                                                             |
| Proposed Solution |                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Approach          | Develop a Machine Learning-based prediction model using the <b>Random Forest algorithm</b> . The model analyzes applicant details such as income, employment, education, family status, credit history, and other relevant features to predict whether the credit card application is likely to be approved or rejected. The trained model is integrated into a Flask web application for real-time predictions. |
| Key Features      | <ul style="list-style-type: none"> <li>Machine Learning-based credit card approval prediction using Random Forest.</li> </ul>                                                                                                                                                                                                                                                                                    |

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|  | <ul style="list-style-type: none"> <li>• Web application developed using Flask for real-time prediction.</li> <li>• User-friendly interface for entering applicant details.</li> <li>• Fast and consistent prediction results.</li> <li>• Model trained using historical applicant and credit history data.</li> </ul> |
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## Resource Requirements

| Resource Type           | Description                             | Specification/Allocation                          |
|-------------------------|-----------------------------------------|---------------------------------------------------|
| <b>Hardware</b>         |                                         |                                                   |
| Computing Resources     | CPU/GPU specifications, number of cores | T4 GPU                                            |
| Memory                  | RAM specifications                      | 8 GB                                              |
| Storage                 | Disk space for data, models, and logs   | 1 TB SSD                                          |
| <b>Software</b>         |                                         |                                                   |
| Frameworks              | Python frameworks                       | Flask                                             |
| Libraries               | Additional libraries                    | scikit-learn, pandas, numpy, matplotlib, seaborn  |
| Development Environment | IDE                                     | Jupyter Notebook, pycharm                         |
| <b>Data</b>             |                                         |                                                   |
| Data                    | Source, size, format                    | Kaggle dataset, 614, csv<br>UCI dataset, 690, csv |